Modelling of AgrA inhibitors to combat anti-microbial resistance in Staphylococcus aureus

Amitha Joy [1], Febin Seethi V [3], Marria C Cyriac[1], Jasmin Habeeb[2], Sunisha Sudhakaran[3], Shaheen Shah[3]

[1]Department of Biotechnology, Sahrdaya College of Engineering and Technology, Thrissur 680684, Kerala, India

[2]Division of Crop Improvement, ICAR-Central Plantation Crops Research Institute, Kasaragod, 671124, Kerala, India[3]Genomics Central [MaGenomics], Thrissur 680125, Kerala, India

ABSTRACT

Staphylococcus aureus is a Gram-positive bacterium found on human skin that causes skin and soft tissue infections, as well as pneumonia, osteomyelitis, and endocarditis. The prevalence of antibiotic resistant strains has made the treatments less effective. An efficient alternate method of battling these contagious diseases is anti-virulence strategy. The AgrA protein, a key activator of S. aureus's Accessory Gene Regulator system, is vital to the organism's virulence and, consequently, its pathogenesis. Using a Machine Learning algorithm, the Support Vector Machine (SVM), and a ligand-based pharmacophore modelling method, prediction models of AgrA inhibitors were developed. The metrics of the SVM model were inadequate, hence it was not used for virtual screening. For ligand-based pharmacophore modelling, 14 of 29 compounds were removed from the active set due to a lack of shared pharmacophore properties, and 504 compounds were designated as decoys. A 3D pharmacophore model was created using LigandScout 4.4.5, with a fit score of 57.48, including a positive ionizable group, one hydrogen bond donor, and three hydrogen bond acceptors. The model after further validation was used to virtually screen an external database which resulted in six hits. These compounds were docked with the AgrA domain crystal structure to determine the inhibitor activity. Further, each docked complex was subjected to a 100 ns molecular dynamics simulation. CID238 and CID20510252 demonstrated potent inhibitory binding interactions and hence can be used to develop AgrA inhibitors in future after proper validation.

Key words: Anti-microbial resistance, Staphylococcus aureus, quorum sensing, anti-virulence, ligand-based pharmacophore model, support vector machine model, machine learning

1. INTRODUCTION

Antibiotic resistant bacterial infections have become a global health issue of major concern [1]. Among the antibiotic resistant bacteria, methicillin resistant *Staphylococcus aureus* (MRSA) poses a major threat to global health [2]. *Staphylococcus aureus* is a Gram-positive bacterium. Infections caused by this bacterium range from skin lesions to osteomyelitis and endocarditis. MRSA strains cause nosocomial infections with high mortality. Treatment of these infections are a challenge due to the antibiotic resistance posed by the bacterium [3].

Most of the pathogenic bacteria produce diverse virulence factors to establish colonisation and infection [2,4]. Hence, inhibition of bacterial signaling systems or virulence mechanisms is a suitable approach for combating antibiotic resistant bacterial infections [1].

Some of the benefits of this approach over conventional antibacterial strategies include reduction of overexposure to antibiotic compounds, targeted disabling of the infection process, better vulnerability of the pathogen to host immune responses, less selective pressure for development of antibiotic resistance and little or no impact on the normal human microbiota [2,4].

The most extensively characterized signal transduction system in *S.aureus* is the Accessory Gene Regulator (*agr*) System. It controls the expression of virulence determinants such as toxins, exoenzymes and surface proteins [2]. The *agr*-system plays a critical role in the infections caused by *S.aureus* [5] and hence, represents a potential anti-virulence drug target [6].

Within the *agr*-system, AgrA can serve as a potential drug target for inhibition of *agr* quorum sensing. The AgrA gene is important for *S. aureus* quorum sensing and survival [7]. The phosphorylation of ASP59 on the N-terminal domain of AgrA, a response regulator protein, is a significant step in the virulence factor production. AgrA, after phosphorylation, forms a dimer and its DNA binding domains at the C-terminal bind with promoter P2. This activates the transcription of autoinducing peptide (AIP). AgrA also binds to P3, the 10-fold weaker binding promoter, when the concentration of AIP reaches a particular threshold. This elicits the expression of a series of virulence factors in *S. aureus* [8]. The DNA binding domain (LytTR domain) binds with the cognate DNA leading to the regulation of gene expression and subsequent synthesis of exotoxin [7]. The co-crystallized structure of AgrA DNA binding domain in complex with a DNA pentadecamer duplex has been reported. $AgrA_c$ (C-terminal DNA binding domain of AgrA) constitutes residues 137 to 238 among which HIS169, ASN201 and ARG233 make distinct contacts with the DNA complex. Alanine mutagenesis and isothermal titration calorimetry studies validated the significance of HIS169 and ARG233 in DNA binding [9]. Inhibiting the binding of phosphorylated AgrA to the P2 and P3 promoters diminishes the formation of virulence factors [10].The amino acid sequences of AgrA are highly conserved compared to AgrB , AgrC or AgrD [11] making it a better target for virulence factor deployment.

Although promising lead compounds have been screened from synthetic and natural product libraries, no *agr*-system inhibitors have made way to the clinic to date [5].

Bioactive compounds are considered to be rich sources of *agr*-system inhibitors [6]. For example, two novel depsipeptides namely solonamide A and B that interfere with the activity of *agr*-system in *S.aureus* were identified from a marine microbe, *Photobacterium* [12]. Similarly, extracts from several Italian medicinal plants have been identified to possess anti-virulence activity against *S.aureus* [13]. In comparison with synthetic entities, natural products are less toxic and possess diverse features and complex structures which may account for different mechanisms of action [14]. Thus, there is a huge demand for natural products with efficient inhibitory activity against *agr*-system and specifically AgrA to combat MRSA infections.

In silico strategies are an emerging approach to screen huge libraries of natural compounds with anti-virulence activity. These bioinformatics tools reduce the time and money invested in pre-clinical phases [15]. Virtual screening helps in identifying novel entities that possess high probability of target protein binding [16]. This approach is cost effective and faster [17].

In the present study, two computational predictive modelling approaches were chosen namely Support Vector Machine, which is a type of machine learning method [18], and ligand-based pharmacophore modelling [19]. Pharmacophore modelling using LigandScout performed better and was employed to virtually screen compounds with potential AgrA inhibiting activity. The screened compounds were further docked with the AgrA protein and a molecular dynamics simulation was also performed to arrive at compounds that had the best inhibitory activity against AgrA protein using in-silico approaches.

2. METHODOLOGY

AgrA inhibitors were collected from scientific publications in PubMed (https://pubmed.ncbi.nlm.nih.gov) by searching with keywords such as 'AgrA', 'quorum sensing', 'inhibit', and 'Staphylococcus aureus' and also from BindingDB database (http://www.bindingdb.org) [20].

2.1 SVM modelling in BindingDB

The inbuilt virtual screening tool in BindingDB was used to develop the SVM model. The structure data file (sdf) files of the inhibitors collected from literature and BindingDB database were combined and served as input for the SVM model. The known actives (inhibitors) were divided into training and test sets. Each set was supplemented with 500 decoy compounds and the descriptors were computed for all compounds. A Support Vector Machine (SVM) model was trained with the training set and applied to the test set and was evaluated according to its ability to enrich known actives in both the training and the test sets.

Since the SVM model trained with the whole set of 29 inhibitors gave low Enrichment Factors (EF), different combinations were tried based on structural similarities. The model trained with a set of 19 compounds gave the best EF.

2.2 Ligand based pharmacophore model

Ligand based pharmacophore modelling was carried out for the selected compounds using LigandScout 4.4.5 [31]. Twenty nine compounds were selected for the initial study. Among the twenty nine compounds, fourteen compounds were removed since there were very few shared pharmacophore features in common. Hence the remaining fourteen compounds were used for building the ligand based pharmacophore model. Pharmacophore fit scoring function and 'Shared feature pharmacophore' was the pharmacophore type used. The bond distance and bond angle between each feature were analyzed.

2.3 Validation of the pharmacophore model and virtual screening

The model was validated using an active set comprising of the same fourteen compounds. The inactive set comprised 504 compounds generated from DUD-E server (http://dude.docking.org/generate) [32]. The active set database and inactive set database were generated in LigandScout in .ldb format. The selected twenty nine AgrA inhibitors from literature are shown in Table 1 and the detailed chemical structures of these compounds are shown in Table S1. A database with 2432 structures from PubChem [33] was generated in .ldb format using LigandScout. This database was virtually screened using the validated model in order to identify new compounds with similar pharmacophore features.

Table 1. The information obtained from the PubChem database for the compounds derived from literature

Compound name	Chemical formula	CAS number	Compound ID (CID) number
Flavone	$C_{15}H_{10}O_2$	525-82-6	CID10680
4',5,7 Trihydroxy_isoflavone (Genistein)	$C_{15}H_{10}O_5$	446-72-0	CID5280961
Morin	$C_{15}H_{10}O_7$	480-16-0	CID5281670
ω-Hydroxyemodin (Citreorosein)	$C_{15}H_{10}O_6$	481-73-2	CID361512
Apicidin	$C_{34}H_{49}N_5O_6$	183506-66-3	CID6918328
Norlichexanthone	$C_{14}H_{10}O_5$	20716-98-7	CID5281657
Savirin	$C_{18}H_{16}N_4O_3S$	-	CID3243271
9H-xanthene-9- carboxylic acid	$C_{22}H_{27}NO_3$	13347-41-6	CID83364
2-(4-methylphenyl)-1,3- thiazole-4-carboxylic acid	$C_{11}H_9NO_2S$	17228-99-8	CID2794795
4-phenoxyphenol	$C_{12}H_{10}O_2$	28515-80-2	CID13254
4-hydroxy-2,6-dimethylbenzonitrile	C_9H_9NO	58537-99-8	CID590183
[5-(2-thienyl)-3-isoxazolyl]methanol	$C_8H_7NO_2S$	194491-44-6	CID2776547
5-acetyl-4-methyl-2-(3-pyridyl) thia-zole (AMPT)	$C_{11}H_{10}N_2OS$	122229-18-9	CID1485005
CHEMBL3289785	$C_{38}H_{58}N_{10}O_{12}S$	-	CID10191219
TURNAGAINOLIDE A	$C_{30}H_{44}N_4O_6$	-	CID53355936
CHEMBL8483	$C_{16}H_{27}NO_4$	168982-69-2	CID3246941
CHEMBL3289786	$C_{17}H_{14}O_6$	-	CID51537215

Emodin	$C_{15}H_{10}O_5$	518-82-1	CID3220
CHEMBL3289787	$C_{17}H_{12}O_6$	-	CID90644759
CHEMBL3289788	$C_{17}H_{14}O_7$	-	CID38357209
CHEMBL3289789	$C_{17}H_{12}O_7$	-	CID90644760
CHEMBL3289790	$C_{15}H_8O_8$	-	CID90644761
Emodic Acid	$C_{15}H_8O_7$	478-45-5	CID361510
2-chloroemodic acid	$C_{15}H_7ClO_7$	-	CID38350655
CHEMBL3742171 (Arthroamide)	$C_{29}H_{42}N_4O_6$	-	CID127038156
CHEMBL3740058	$C_{54}H_{68}N_8O_{13}$	-	CID6918144
CHEMBL3741390	$C_{34}H_{37}N_5O_5$	-	CID127038497
Bumetanide	$C_{17}H_{20}N_2O_5S$	28395-03-1	CID2471
Diflunisal	$C_{13}H_8F_2O_3$	22494-42-4	CID3059

2.4 Interaction studies

The hits obtained from virtual screening were chosen for the docking study using AutoDock Vina [34]. The crystal structure of AgrA LytTR domain, labeled as 4XQJ in the Protein Data Bank (PDB), was chosen as the protein target and it was downloaded from the PDB website. With a resolution of 1.9 A^0 and all of the residues in the allowed regions of the Ramachandran plot, the crystal structure, with the PDB ID 4XQJ, is a high-quality structure. The binding cavities were predicted using Discovery Studio (DS) Visualizer [35].. Counter maps with x, y, z coordinates of the best binding pocket with corresponding amino acids were obtained as 13.506, 6.257 and 48.055 respectively. The ligands were prepared as pdbqt file using OpenBabel [36]. The configuration file was generated using the binding site co-ordinates. The protein structure was prepared by adding with Kollman charges [37] and polar hydrogen atoms. The heteromolecules like ethanol, calcium ions and 1, 2 ethanediol were removed. Water molecules were removed from the protein during protein preparation so that the binding pocket is cleared from possible water molecules that would affect the pose search. The prepared protein structure was saved as pdbqt file. The exhaustiveness was set as 8 and the grid box size of 25 was set for x, y and z dimensions. The output files were also generated in pdbqt format. The protein structure was docked with all the six hits.

2.5 Molecular dynamics simulation studies

Molecular dynamics (MD) simulation studies aim to understand the dynamic properties of the molecules [38]. The simulation studies help to check the stability of the structures. MD studies were carried out using Desmond which uses OPLS forcefield [39]. Proteinprep wizard of Desmond was used for preprocessing and optimizing the AgrA protein. The preprocessing involves assigning of bond order, addition of hydrogen atoms; metals with zero-order bonds and disulfide bonds were created. SPC water model was used for building the MD system in a cubic boundary box with a distance of $10A^0$ and the volume was minimized. The system was neutralized using Cl- counter ions and minimized the energy. This was followed by energy minimization step. A temperature of 300K and pressure 1 bar, ensemble isotherm-isobar (NPT) was selected and simulation of AgrA protein was carried out for 100 nanoseconds. The same conditions were retained for running the simulations of docked complexes.

3. RESULTS AND DISCUSSION

3.1 Support Vector Machine model

The metrics of the SVM model with a set of 19 compounds are given in Table 2.

Table 2. SVM model metrics

	Training set (n = 10)	Test set (n = 9)
EF (2%):	51	45.24
EF (10%):	10	9.05

Support vector machine virtual screening method is known to be a rapid computational tool for predicting potential drug candidates with good accuracy and specificity which can efficiently discriminate inhibitors and non-inhibitors [40]. The enrichment factors at 2% and 10% were 45.24 and 9.05 respectively in the test set. In the current study, we understand that due to the limited number of known AgrA inhibitors and structural differences among them resulted in a model with low performance.

3.2 Pharmacophore model

In Ligand-based 3D pharmacophore model generation and model validation study, pharmacophore models based on the Training set (14 active inhibitors) and Inactive set (504 compounds) were created using LigandScout (www.inteligand.com/ligandscout) [31]. The pharmacophore model generated is as shown in Figure 1. The bond length and bond angle between each feature is also depicted in the Figure 1. Each colour indicates a specific pharmacophore feature.. The pharmacophore fit score was obtained as 57.48, and the five features, (three Hydrogen Bond Acceptors (HBA) (red sphere), one Hydrogen Bond Donor (HBD) (green sphere), and one positive ionizable group (blue sphere)), depicted in Figure 1 were present for all the compounds, this color coding is similar as followed earlier by Seidel et al. [41].

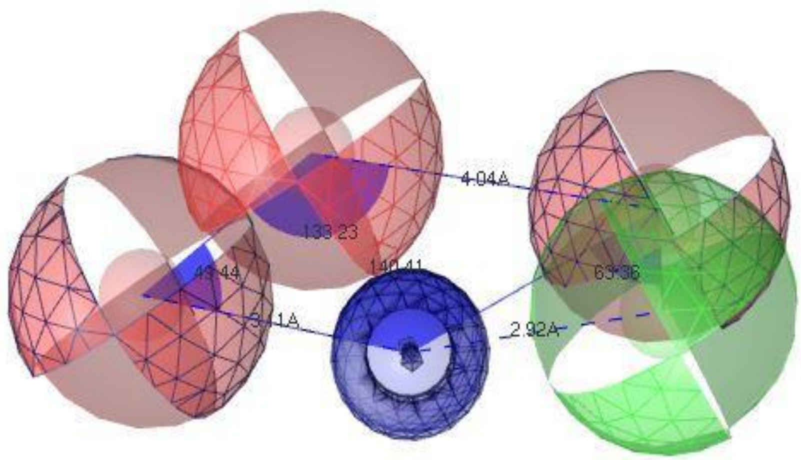

Figure 1. Pharmacophore model based on 14 active ligands

In order to validate the ligand-based 3D pharmacophore model, 14 active molecules taken from literature and 504 decoys compounds from the DUD-E server were selected. In the receiver operating characteristic curve (ROC) of the generated model, the true positive rate (rate of recovering actives) is represented by the ROC curve's Y-coordinate, and the acceptable false positive rate (rate of retrieving inactive or decoys) is indicated by the ROC curve's X-coordinate (Figure 2) [42]. There were no false positive hits and the model correctly predicted 12 of the 14 actives to be true positives. With a good area under the ROC curve (AUC) value of 1.00 and an early enrichment factor (EF1%) of 36.8, the pharmacophore model was successfully able to identify true active from decoy, as shown in Figure [43]. With a sensitivity of around 86% and an AUC value of 93.0%, 12 Hits were discovered. The PubChem database of 2432 compounds was used for virtual screening to further validate the predicted model. This allows for the identification of unique, diverse hits, and 6 hits were obtained (Table 3), validating the proposed model as the best-predicted model.

3.3 Validation of the pharmacophore model

The model predicted 12 actives out of the 14 as true positives and there were no false positive hits. The Receiver operating characteristic curve (ROC) obtained is given in Figure 2. The pharmacophore model when screened against 2432 compounds downloaded from Pubchem repository gave 6 hits as shown in Table 3. The resulting hits show similar pharmacophore characteristics to the validated model's pharmacophore features.

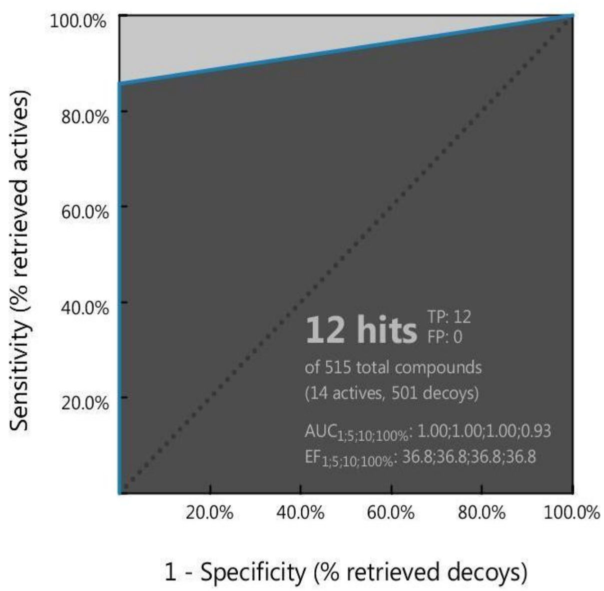

Figure 2. ROC curve of pharmacophore model

The real positive rate (retrieval frequency of actives) is shown by the Y-coordinate of the ROC curve, while the appropriate false positive rate is indicated by the X-coordinate (retrieval frequency of decoys).

Table 3. The information obtained from the PubChem database for the compounds examined in our docking study

CID number	Chemical formula	Status (Active/Decoy)	Pharmacophore Fit score
CID238918	$C_{16}H_{18}N_8O_9$	active	57.31

CID238	$C_{10}H_{16}N_5O_{13}P_3$	active	56.63
CID13132413	$C_{13}H_{19}FN_2O_6$	active	56.34
CID549884	$C_7H_7F_3N_4O_3$	active	56.3
CID54173680	$C_9H_9N_8^+$	active	56.2
CID20510252	$C_{14}H_{20}N_6O$	active	55.73

3.4 Docking results

The crystal structure of AgrA LytTR domain in complex with promoters was chosen as the protein target and was retrieved from RCSB (PDB ID: 4XQJ) with a resolution of 1.90 Å. The determination of the best conformation based on Binding Energy scores [44] was made possible by the molecular docking of six hits, as shown in Table 4. The shape of docked protein-ligand complex reveals interactions with hydrogen bonds, hydrophobic forces, and non-bonded forces are shown in Figure 3a and 3b. The screened 6 hits had binding energies against AgrA that ranged from -8.7 to -6.6kcal/mol.

Table 4. The binding energies of the compounds examined in our docking study

CID number	Chemical formula	Binding Energy (KJ/mol)
CID238918	$C_{16}H_{18}N_8O_9$	-8.7
CID238	$C_{10}H_{16}N_5O_{13}P_3$	-8.1
CID54173680	$C_9H_9N_8^+$	-7.6
CID549884	$C_7H_7F_3N_4O_3$	-7.6
CID20510252	$C_{14}H_{20}N_6O$	-7.0
CID13132413	$C_{13}H_{19}FN_2O_6$	-6.6

The binding affinity of compound CID238 against AgrA was -8.1kcal/mol. The interactions revealed four conventional hydrogen bonds with residues, including GLU163, SER164, SER168, and SER202, as well as carbon hydrogen bond with SER168. Compound CID238 additionally exhibits one hydrophobic interaction (pi-Sigma) with SER168, and van der Waals interactions with SER165, THR166, LYS167, HIS169, PHE203, HIS227, and TYR229 were also observed. Furthermore, the attractive charge was determined for the residues GLU163 with AgrA and shown in Figure 3a.

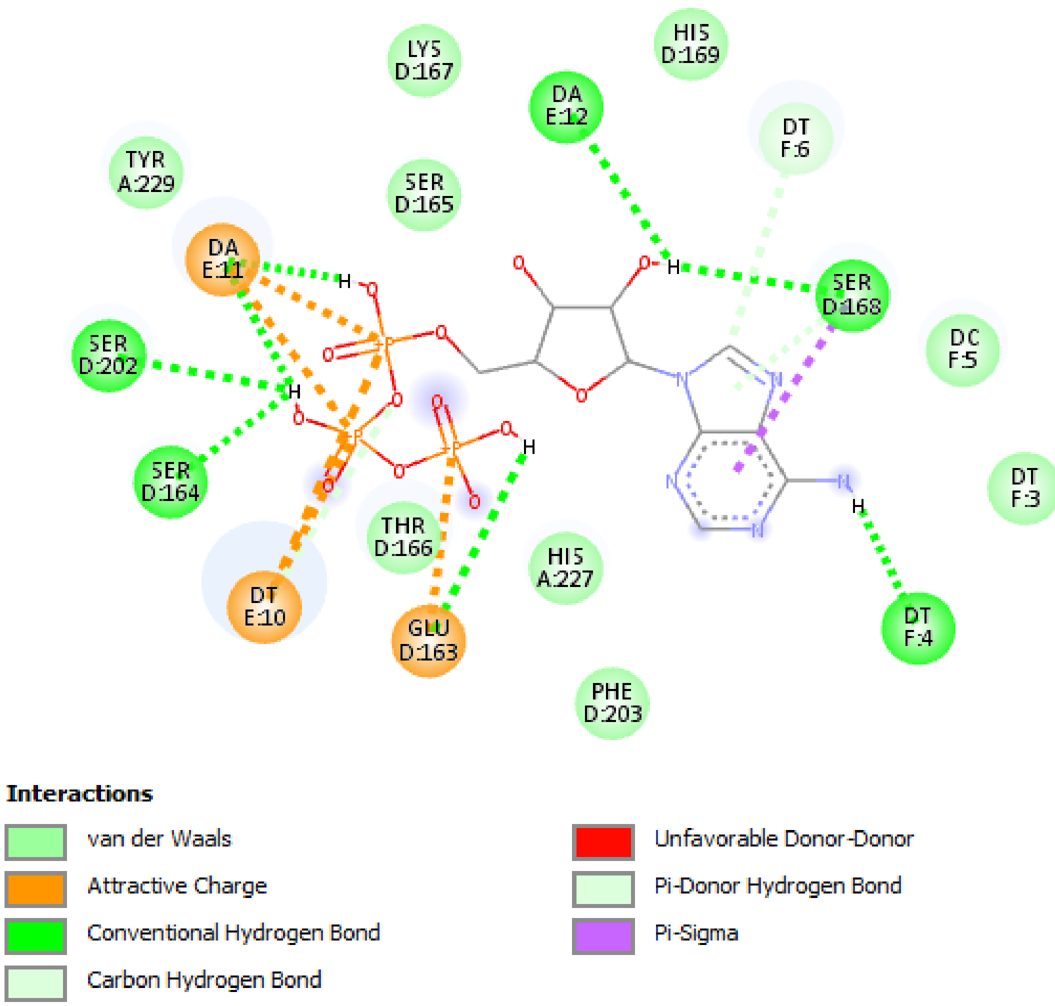

Figure 3a. Interactions between the AgrA protein (PDB ID: 4XQJ) and the ligand molecule (CID 238)

The binding affinity of compound CID20510252 for AgrA was -7.0kcal/mol. Our analyses revealed four typical hydrogen bonds with residues, including those with SER165, and HIS227. Furthermore, compound CID20510252 exhibits van der Waals contacts with residues SER168, and HIS227, shown in Figure 3b. According to the findings of the docking study, seven residues were identified as active AgrA binding sites (GLU163, SER164, SER165, SER168, HIS169, SER202 and TYR229).

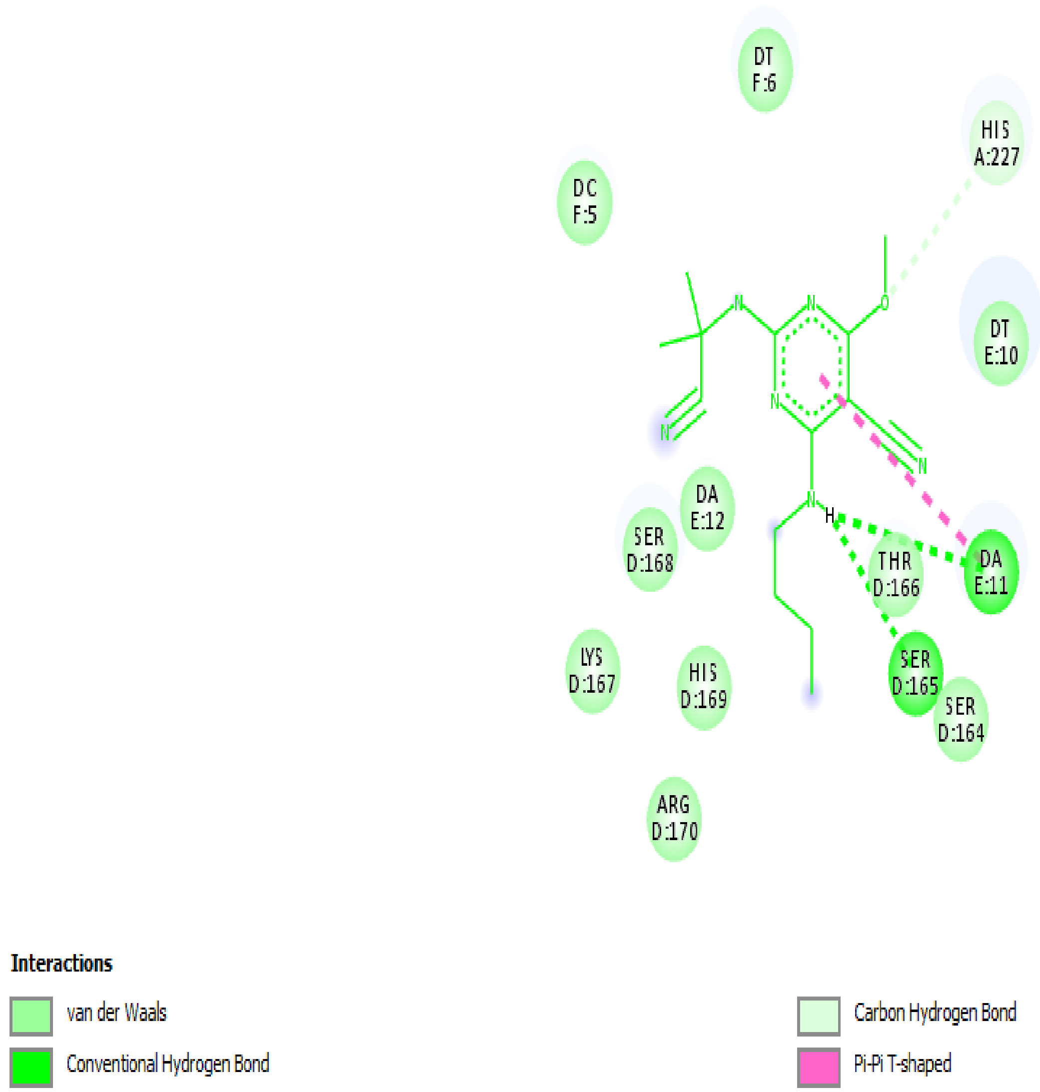

Figure 3b. Interactions between the AgrA protein (PDB ID: 4XQJ) and the ligand molecule (CID20510252)

3.5 Molecular dynamics simulation

The simulation studies of the target AgrA protein (PDB ID: 4XQJ) exhibited root mean square deviation (RMSD) 1.724A° in the trajectory analysis whereas the compound ID 238 with RMSD 2.33 A° and for the compound ID 20510252 with RMSD 2.02 A° and the system got converged at and was stable as shown in Figure 4. Rest of the compounds failed in system building step.

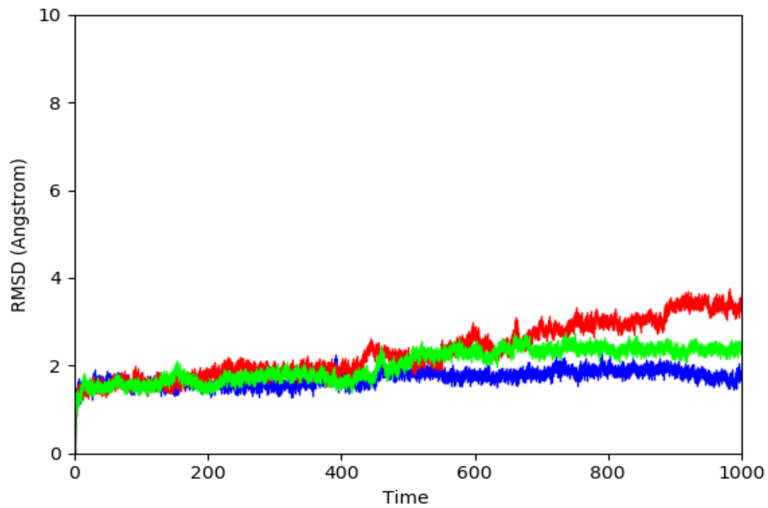

Figure 4. The RMSD data calculated for the structure of the AgrA protein (PDB ID: 4XQJ) and the ligand molecule. While the blue curve shows the RMSD data of the AgrA protein, the green and red curves show the RMSD data of the ligand molecules (CID 20510252 and CID238) respectively

The obtained root mean square fluctuation (RMSF) values showed that the AgrA protein (PDB ID: 4XQJ) remained stable throughout the simulation. This is also true for the examined ligand molecules (CIDs 238 and 20510252) as shown in Figure 5.

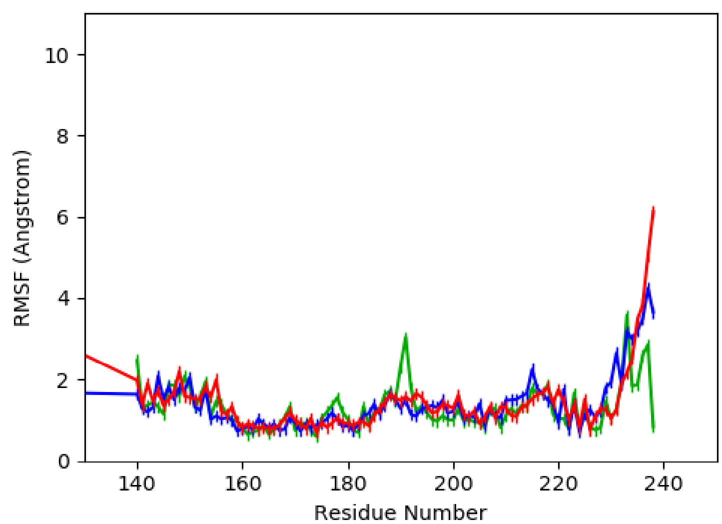

Figure 5. The RMSF data calculated for the structure of the AgrA protein (PDB ID: 4XQJ) and the ligand molecule. While the blue curve shows the RMSF data of the AgrA protein, the green and red curves show the RMSF data of the ligand molecules (CID 20510252 and CID238) respectively

The radius of gyration of the AgrA protein (PDB ID: 4XQJ) was found within the range of 17.757 Å. As the protein is docked by a ligand, it decreases notably; the corresponding radius values were found 17.745 Å for the ligand CID 238 and 17.69 Å for the ligand CID 20510252.

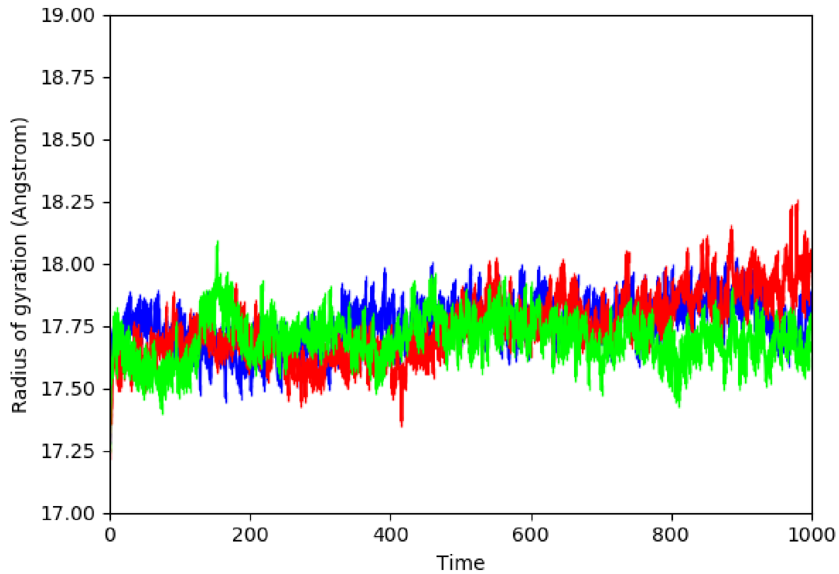

Figure 6. The Radius of gyration data calculated for the structure of the AgrA protein (PDB ID: 4XQJ) and the ligand molecule. While the blue curve shows the radius of gyration data of the AgrA protein, the green and red curves show the radius of gyration data of the ligand molecules (CID 20510252 and CID238) respectively

Figure 7 shows the ligand-protein interactions that occur more than 30% of the time throughout the simulation of the CID238 protein complex. HIS169 is the only residue which occur more than 30% of the time throughout the simulation of the CID238 protein complex.

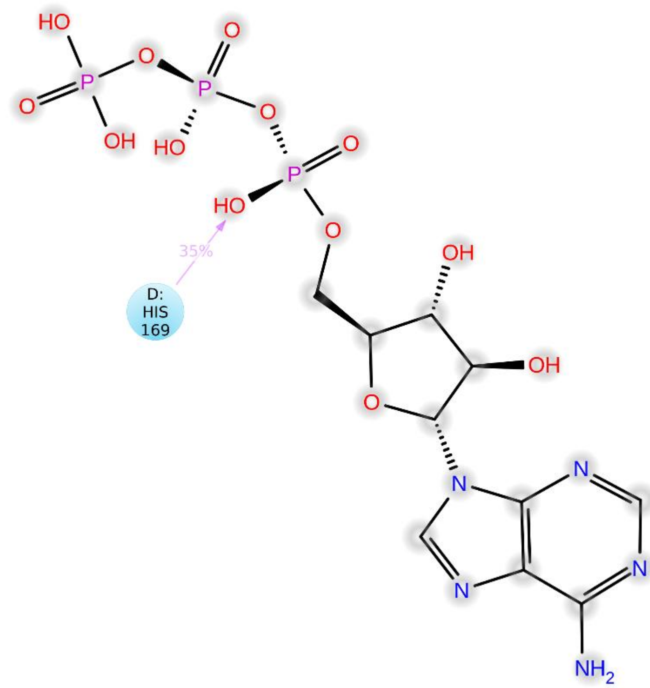

Figure 7. Interaction between AgrA protein and CID238

By computing three properties- root mean square deviation (RMSD), root mean square fluctuation (RMSF) and Radius of gyration (Rg), during a 100ns simulation, the structural and conformational stability of the protein-ligand complex was examined [38]. The protein's RMSD maintained 1.72 Å until the simulation's end. The RMSD calculated for the ligand CID238 exhibited a change in 1.72 - 2.33 Å range in the simulation time interval between 10 and 42 ns, and then it reached the value 3.8 Å at the end of the simulation. The ligand CID 20510252 remained stable by yielding a RMSD value in range 1.72 -2.02 Å during the time interval between 10 and 48 ns. Then, it reached the value 2.20 Å between 48.1 and 49 ns, and remained steady with a similar RMDS in 2.20 -2.30 Å range until the end of the simulation. Among the simulated three structures, the ones with the ligands CID 20510252 and CID 238 exhibited the lowest RMSD profile and hence the greatest stability during the simulation (Figure 4). Similarly, the ligand CID 20510252 and CID 238 complexes have fewer residue fluctuations than that observed for the protein 4XQJ (see Figure 5). In order to evaluate the structural stability of proteins, the "radius of gyration (Rg)" value assesses how compact a protein is. Here, during the experiment, the Rg of the structures exhibited a significant fluctuation between 17.27 and 18.25 Å. As shown in Figure 6, the protein's radius of gyration value was determined to be about 17.757 Å. This value is notably closer to those determined for the structures involving the ligand CID 238 (about 17.74 Å) or the ligand CID

20510252 (about 17.69 Å) when compared to those determined for the other examined ligand complexes. Collectively, these findings have suggested that the target protein (AgrA) and the ligand CID 20510252 are to create a stable complex when they form a favorable contact through the key amino acid residues.

4. CONCLUSION

The accessory gene regulator (*agr)* system plays a critical role in the infections caused by *S.aureus* and hence, represents a potential anti-virulence drug target. For the treatment of MRSA infections, there is a high demand for natural compounds having potent inhibitory activity against the *agr*-system, and more specifically AgrA. Despite screening potential lead compounds from synthetic and natural product libraries, no *agr*-system inhibitors have been discovered so far. In the current work, Support Vector Machine and ligand-based pharmacophore modelling were chosen as the two computational predictive modelling strategies. The validated ligand based pharmacophore model fetched six compounds from Pubchem database. The two best hits, CID20510252 and CID238, were found after the collected hits were screened using docking and molecular dynamics simulation. These two potent inhibitory binding interactions can be used to develop AgrA inhibitors in future. In conclusion, these theoretical results indicate that more research studies on the natural products are needed to identify bioactive compounds in a bid to strengthen, diversify and sustain the fight against the anti-microbial resistance.

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